

Predefined Methods

There are predefined methods supplied with BioFocus operating software. These methods can be used without modification. You must, however, ensure that the proper vials are in place in the inlet and outlet carousels.

SHUTDOWN Method

The Shutdown method template shown in Figure 5-77 uses the Turn Lamps Off option, extending the detector lamp life if the automation sequence finishes many hours before you return to the lab. It also cleans and dries the capillary after the last run with four capillary purges: Water, Cap_Wash, Water, and Nitrogen gas. We recommend routinely making it the last run group in an automation sequence.

Define Method

Method ID: **SHUTDOWN** OK Inlet Reagent: **WATER**

Description: **terminates autosequence** Cancel Outlet Reagent: **WATER**

Run Parameters:

⚙ Voltage Mode Run Voltage: **0.0** kV Polarity: **+** **+** **+** **-** **-** **+** **+** **+**

⚙ Current Mode Current Limit: **0.3** μA Capillary Temperature: **20** °C

Run Time: **1.00** minute[s]

Preparation Cycle Parameters: 4 cycle(s) defined

Cycle	Pre/Post	Inlet	Outlet	Pressure	Duration	Units
1	Pre-inj	WATER	Waste	High	60	sec
2	Pre-inj	CAP_WASH	Waste	High	60	sec
3	Pre-inj	WATER	Waste	High	60	sec

Add Insert Delete Delete All

Injection: **None**

Detector: **Single Wavelength Mode, 200 nm**

Timed Events

Figure 5-77 Shutdown Method Template

Be sure that you have assigned WATER, CAP_WASH, and NITROGEN (insert an empty vial for gas) positions in the inlet and one WATER position in the outlet carousel of the selected configuration. Although no sample will be run, you need to assign a sample and WATER as Inlet and Outlet Reagents.

The Shutdown method template is not editable. You can define a new (editable) Shutdown method using the template by clicking the **Define** button in the Browse Methods screen.



Vials Up Button

If you prefer that your capillary tips be immersed in a neutral solution at the completion of an automation sequence (to avoid capillary plugging or desiccation of a capillary gel), be sure to define the new Shutdown Method with this solution as the Inlet and Outlet Reagents. Then click the **VIALS UP** button on the Main Screen Toolbar (or select it under the Commands menu) before starting an automation sequence.

PCR Product/SDS-Protein Methods

In addition to CZE, IEF and Shutdown, predefined method templates for analyzing PCR products and SDS-complexed proteins using the Bio-Rad CE kit chemistries are supplied in the database. These method templates are noneditable and nonautodeletable but are singly deletable only after deletion of the automation sequence(s) to which they are linked.

To view the details of a template method, double-click the desired template in the Browse Methods screen. This will bring up the Define Method screen for the selected template.

The Define Method screen shown in Figure 5-78 illustrates run parameters for the ds1000 method template. These run conditions are recommended for analyzing double-stranded DNA from 100 to 1000 bp on the BioFocus system using Bio-Rad's ds1000 Analysis Kit reagents with a special Bio-Rad coated capillary (24 cm x 20 μ m).

The screenshot shows the 'Define Method' dialog box for the 'ds1000' method. It includes fields for Method ID, Description, Run Parameters (Voltage Mode, Current Mode, Run Voltage, Current Limit, Polarity, Capillary Temperature, Run Time), Preparation Cycle Parameters (a table with 2 cycles), Injection, Detector, and Timed Events.

Preparation Cycle Parameters: 2 cycle(s) defined						
Cycle	Pre/Post	Inlet	Outlet	Pressure	Duration	Units
1	Pre-inj	PCR PREP	Waste	High	180	sec
2	Pre-inj	WATER_1	Waste	High	0	sec

Figure 5-78 PCR Product Method Template

The run parameters for the SDS-Protein method template are shown in Figure 5-79. These conditions are recommended for the analysis of SDS-Protein complexes on the BioFocus system using the Bio-Rad CE SDS-Protein Kit reagents with an uncoated capillary (24 cm x 50 μ m).

Define Method

Method ID: SDS_PROTEIN OK Inlet Reagent: SDS_RUN

Description: SDS Protein method Cancel Outlet Reagent: SDS_RUN

Run Parameters:

• Voltage Mode Run Voltage: 15.0 kV Polarity: + → - Capillary Temperature: 20 °C

• Current Mode Current Limit: 50.0 µA Run Time: 15.00 minute(s)

Preparation Cycle Parameters: 5 cycle(s) defined

Cycle	Pre/Post	Inlet	Outlet	Pressure	Duration	Units
1	Pre-inj	BASE_WASH	Waste	High	90	sec
2	Pre-inj	ACID_WASH	Waste	High	60	sec
3	Pre-inj	SDS_PREP	Waste	High	120	sec

Add Insert Delete Delete All

Injection: Electrophoretic, 10.00 kV, 5 seconds

Detector: Single Wavelength Mode, 220 nm

Timed Events

Figure 5-79 SDS-Protein Method Template

If you wish to use Multi Inject with CE SDS-Protein kit, it is best to inject the internal marker (prepared in water) using a Pre Inject cycle at 120 psi * sec before the electrophoretic injection of the sample. For the Pre Inject cycle, use the internal marker as the Inlet and the run buffer as the Outlet. Make sure to include multiple uniquely-labeled buffers and the internal marker you plan to inject in the reagent database and your Configuration.

You can use the ds1000 or the CE SDS-Protein method template in an automation sequence as is or you can define a new method from the desired template in the Browse Methods screen by clicking the **Define** button. You can then change any of the template parameters to define the new method. When you have finished programming, click **OK** to save your new entry.

Automation Sequences

An Automation sequence links a run Method(s) with a compatible carousel/cartridge Configuration. A Method(s) and a Configuration are compatible only when every reagent assigned by the Method(s) has also been assigned to a vial position in the selected Configuration.

A Run Group is one or more runs performed with one or more samples (a “sample set”) using the same Method. An Automation Sequence is composed of one or more run groups using the same configuration. Each run group can have a different method, as long as each method is compatible with the current configuration.

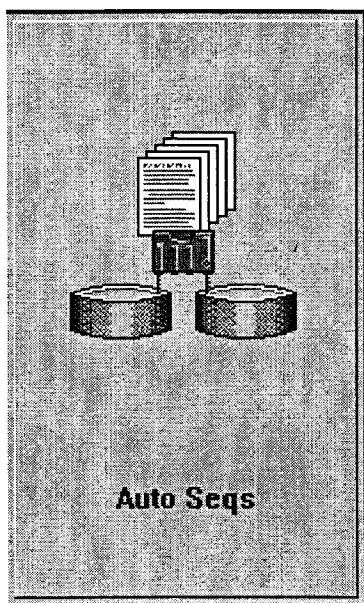
Up to 198 runs or methods (two combined autosequences) can be programmed to run while the instrument is unattended.

Defining New Automation Sequence ID and Selecting Configuration

When you click the **Auto Seqs** button on the main screen, you will see a Browse Automation Sequences screen (you can also access this screen from the menu bar).

NOTE: Only automation sequences corresponding to the current program detection mode (Absorbance or LIF) are available for viewing, defining, editing, or running.

The first time you use the BioFocus program in absorbance mode, this screen will have the five Bio-Rad supplied automation sequence templates shown in Figure 5-80. If you are using LIF detection, the screen will show LIF templates.



Auto Sequences Button

The reference Automation Sequence templates (CZE, IEF, ds1000, Verify, and SDS) link the appropriate reference Method templates with the corresponding reference Configuration templates in the database and can be used to schedule and/or start an automation sequence.

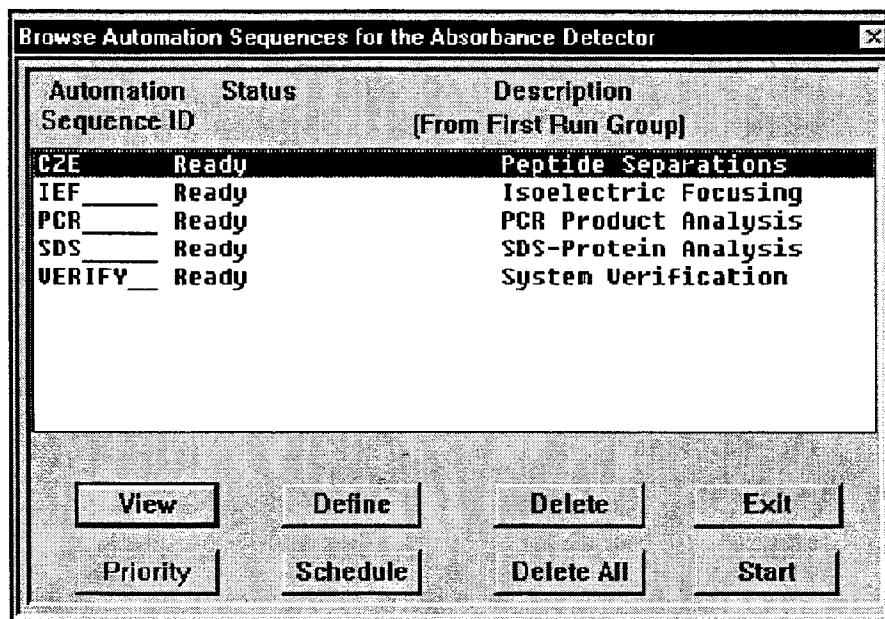


Figure 5-80 Browse Absorbance Automation Sequences Screen

The reference Automation Sequence templates can be deleted singly or automatically, unless they are scheduled, executing, or interrupted.

Defining a New Automation Sequence

You can define a new Automation Sequence using one of the autosequence templates in the database. To define a new automation sequence from a template, highlight the desired template in the Browse Automation Sequences screen and then click the **Define** button (you must have at least one configuration and one method defined). If there are no autosequences in the database, then simply click the **Define** button. For this exercise, the CZE template has been selected.

Assigning an Automation Sequence ID

The Define screen, shown in Figure 5-81, displays an ID for the automation sequence you are about to define. The computer automatically assigns the date (day-month-year) with a letter of the alphabet (A-Z) on each calendar day to each new automation sequence, but you can overwrite this default ID. An automation sequence ID number must be exactly eight characters, with no spaces.

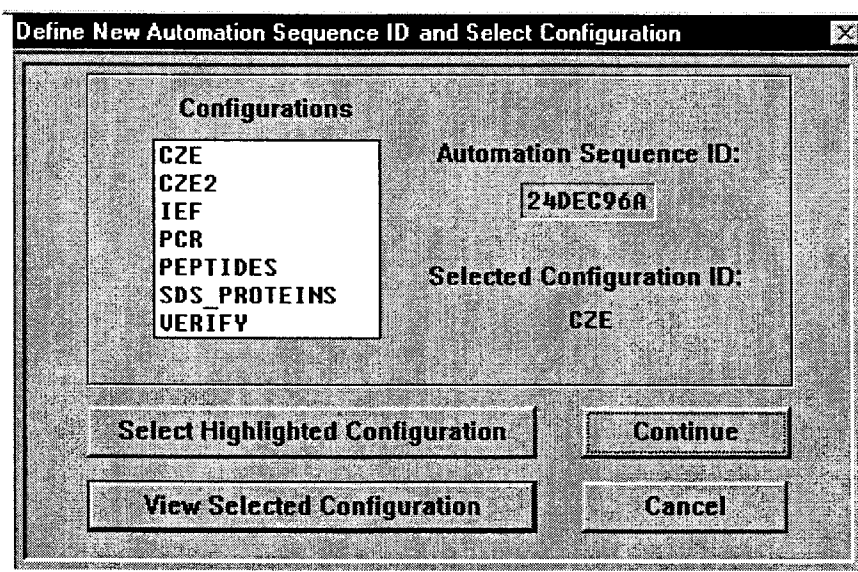


Figure 5-81 Define New Automation Sequence Screen

Selecting Configurations

The Define screen also gives you a listing of the configurations currently in your database library and allows you to view or change the configuration currently recognized by the computer and listed beneath the Selected Configuration ID. If you wish to change it, select the desired configuration from the list by highlighting it, then click the **Select Highlighted Configuration** button (or alternatively, double-click the desired configuration). The new Configuration ID you have chosen should appear beneath the Selected Configuration ID. If you click **Cancel**, you will return to the main screen. If you click **Continue**, the next screen that appears is Define Run Groups for Automation Sequence. Refer to Figure 5-82 .

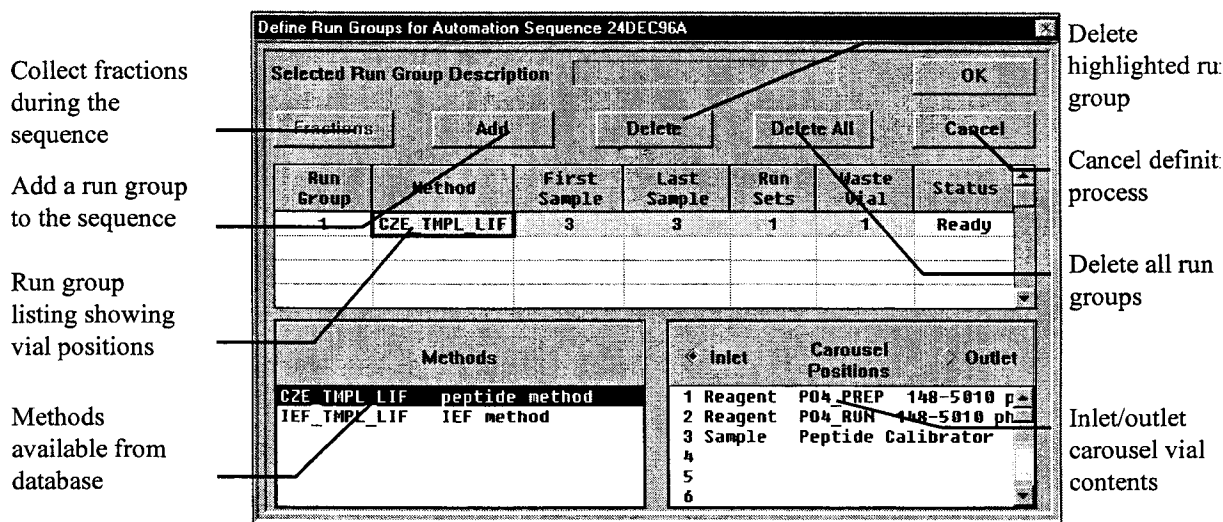


Figure 5-82 Define Run Groups Screen

Creating Run Groups

The Define Run Groups screen (see Figure 5-83) comes up with its template parameters in place. To define a new run group, you can either change each entry in each cell as required or you can delete the entire run group by clicking the **Delete All** button.

Assigning a Method

NOTE: Within the same Run Group, run group related information (Method, First and Last Sample, Run Sets, and Waste Vial) does not have to be entered in any specific order. You can simply click the cell of interest and fill in the appropriate information.

To create a new run group, the first thing you should do is assign a method by double-clicking the desired method in the list box located at the lower left of the screen. The newly selected method (Method 1) will now appear under Method and the neighboring cell (First Sample) will automatically be activated to accept an entry.

Only methods corresponding to the current program detection mode (Absorbance or LIF) are available for assignment.

Placing Samples On the Carousel

Samples to be analyzed within a Run Group must be located in a contiguous series on the inlet carousel. Vial positions for the inlet and outlet carousels appear at the lower right of the screen. The **Inlet** and **Outlet** buttons toggle the view between inlet and outlet carousel positions. To assign the first sample position, first select the Inlet Carousel. Then double-click the desired sample vial in the inlet carousel. This vial number now appears in the First Sample column and the next cell (Last Sample) is automatically activated to accept the new entry. Next, select the Last Sample by double-clicking any sample position that is contiguous with and greater than the First Sample position. This number should now appear in the Last Sample column.

Define Run Groups for Automation Sequence 03FE89ZA

Selected Run Group Description: P-injection

Buttons: Fractions, Add, Delete, Delete All, Cancel, OK

Run Group	Method	First Sample	Last Sample	Run Sets	Waste Vial	Status
1	METHOD1	3	3	1	1	Ready

Methods:

- CZE TEMPLATE peptide method
- EQUILIBRATE Equilibration Met
- FINAL Terminates Autose
- IEF TEMPLATE IEF method
- METHOD1 peptide method
- PCR PCR Product metho

Inlet Carousel Positions:

- 1 Reagent P04 PREP 148-5010
- 2 Reagent P04 RUN 148-5010
- 3 Sample Peptide Calibrator
- 4
- 5
- 6

Figure 5-83 Define Run Groups Screen-New Run Group Created

Assigning Run Sets

Run sets refers to the number of times each sample is run with this method—you can either type in the appropriate number or use the increase/decrease buttons. The default number for run sets is 1. Replicate runs for one sample are performed before advancing to the next sample in the set.

Assigning Waste Vial Positions

Assigned waste position(s) in the outlet carousel appear under the Waste Vial cell. The default position for Waste is the first waste position in the selected configuration.

For long automation sequences with several run groups, or for overnight runs, you may need to assign more than one waste vial. Multiple waste positions should be defined in the configuration selected for this automation sequence. You may have up to 31 entries of waste vials in the outlet carousel, and they do not have to be contiguous.

To assign a different waste location for this run group, double-click the Waste Vial cell. The next screen that you will see is the Waste Locations screen, which lists all waste locations in the configuration template that you have selected. Refer to Figure 5-84.

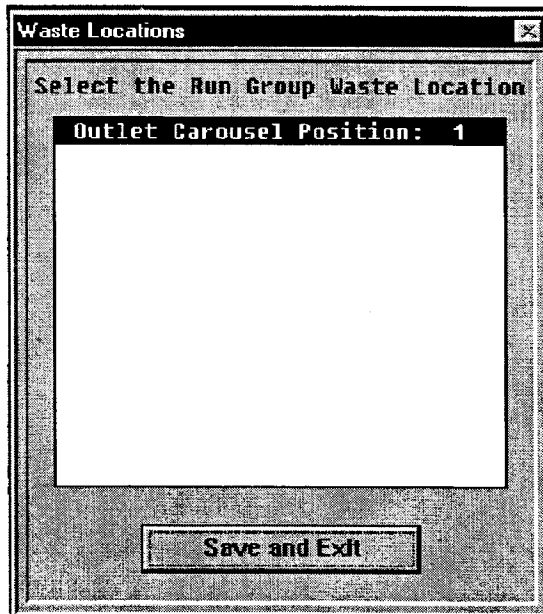


Figure 5-84 Waste Locations Screen

Select a waste location by clicking the desired outlet carousel position and then click the **Save** and **Exit** button at the bottom of the Waste Locations screen to save your entry. This brings you back to the Define Run Groups for Automation Sequence screen shown in Figure 5-85. The newly assigned waste position now appears under Waste Vial.

Collecting Fractions

If the current configuration includes assigned fraction vials in the outlet carousel and you wish to collect fractions during this run group, click **Fractions** and enter the fractions-related information. For details on fraction collection, see “Fractions Collection “ in this section. You may also describe your run group in the Run Group Description box provided. When you have finished defining a **Run Group** (Method, First and Last Sample Positions, Run Sets, Run Group Description, Run Group Waste Vial, and Fractions), click **OK**. If you wish to define a second run group, click **Add**. This changes the first run group’s Status from Pending to Ready.

Define Run Groups for Automation Sequence 24DEC96A

Selected Run Group Description

Fractions Add Delete Delete All Cancel

Run Group	Method	First Sample	Last Sample	Run Sets	Waste Vial	Status
1	IEF_TMPL_LIF	3	3	1	1	Conflict

Methods

CZE_TMPL_LIF	peptide method
IEF_TMPL_LIF	IEF method

Carousel Positions

	Inlet	Outlet
1	Reagent	P04_PREP 148-5010 p
2	Reagent	P04_RUN 148-5010 ph
3	Sample	Peptide Calibrator
4		
5		
6		

OK

Figure 5-85 Define Run Groups Screen-Conflict Indicated In Status Column

Identifying Conflicts

The computer compares the Washes, Reagents, Wastes, Samples, and Fractions in the scheduled configuration with those in the method you have just selected for this run group. Under Status, “Conflict” appears if any substance is called for in a method but is not present in a vial of the current configuration. For detailed information on method or configuration conflicts, click the **Conflict** cell. See Figure 5-86.

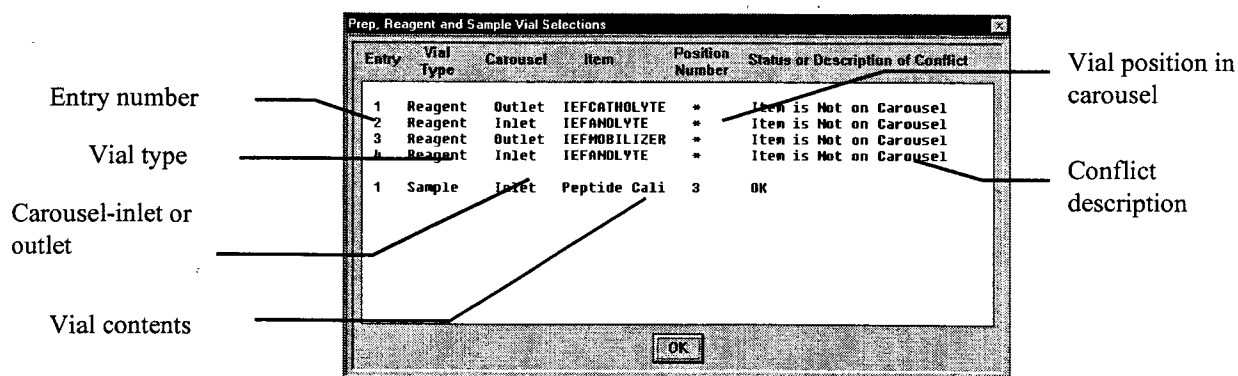


Figure 5-86 Prep, Reagent And Sample Selections Screen

To resolve the conflict, you must either select another method or cancel this automation sequence definition and redefine a compatible configuration. If there is no conflict, you may proceed.

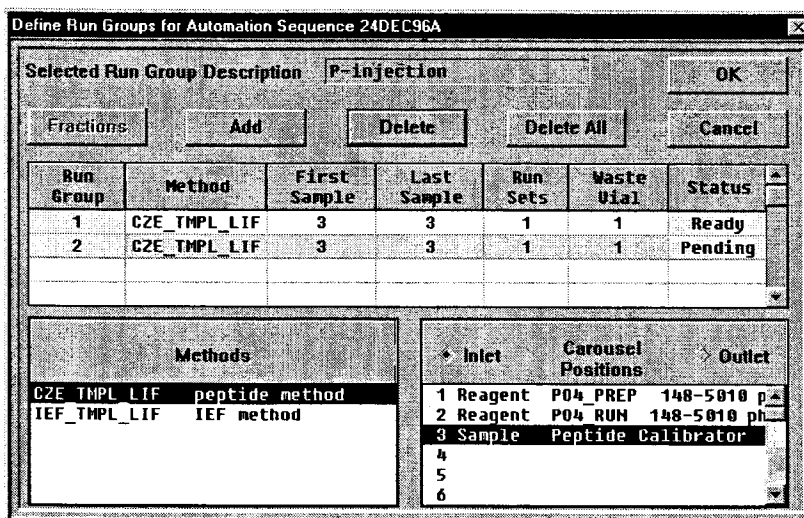


Figure 5-87 Define Run Groups Screen-Second Run Group Added

Defining Additional Run Groups

To add an additional Run Group to the automation sequence, click the **Add** button (or the scroll down arrow or an empty row) at the upper left of the screen shown in Figure 5-87. This selects (highlights) the second row of the Run Group Definitions Table. You can now enter the information for the second run group by repeating all the steps described above. If you wish, you can also assign a new waste position for this Run Group.

If you plan to change to a fresh run buffer vial after a number of runs (to minimize sample contamination of the buffer or to avoid ion depletion), use identical methods with unique labels (described in the Methods section). For example, Run Group 1 may use MethodA for five Run Sets

(uses “BufferA” in the carousels); Run Group 2 may use MethodB for the next five Run Sets (MethodB has the same parameters as MethodA except that it uses “BufferB”, which is simply a fresh vial of the same buffer found in the “BufferA” vial); Run Group 3 may use MethodC for the next five Run Sets, etc.

NOTE: To save a Shutdown Run Group, you need to assign one sample to First and Last Sample positions even though no sample will be injected.

For overnight runs, we strongly recommend a SHUTDOWN Run Group be added as your last run group in your automation sequence. See Figure 5-88.

Run Group	Method	First Sample	Last Sample	Run Sets	Waste Vial	Status
1	CZE_TMPL_LIF	3	3	1	1	Ready
2	SHUTDOWN_LIF	3	3	1	1	Pending

Methods	
CZE_TMPL_LIF	peptide method
IEF_TMPL_LIF	IEF method
SHUTDOWN_LIF	peptide method

	Inlet	Carousel Positions	Outlet
1	Reagent	PD4_PREP 148-5010 p	
2	Reagent	PD4_RUN 148-5010 p	
3	Sample	Peptide Calibrator	
4	Reagent	WATER deionized	
5	Reagent	CAP_WASH 148-5022 c	
6	Reagent	WATER 1 deionized c	

Figure 5-88 Define Run Groups Screen—Showing Shutdown Run Group

Editing a Run Group

If you would like to edit a Run Group which is in Ready status, simply click the cell that you wish to change and enter the new information. After you make the change, the Status of this Run Group becomes Pending.

Deleting Run Groups

Clicking the **Delete** button at the top of the screen deletes the selected run group. The **Delete All** button deletes all run groups in the automation sequence. Clicking the **Cancel** button deletes all run groups in the automation sequence before exiting the Define screen.

When you have finished adding all run groups to your automation sequence, click the **OK** button to go back to the Browse Automation Sequences screen, shown in Figure 5-89.

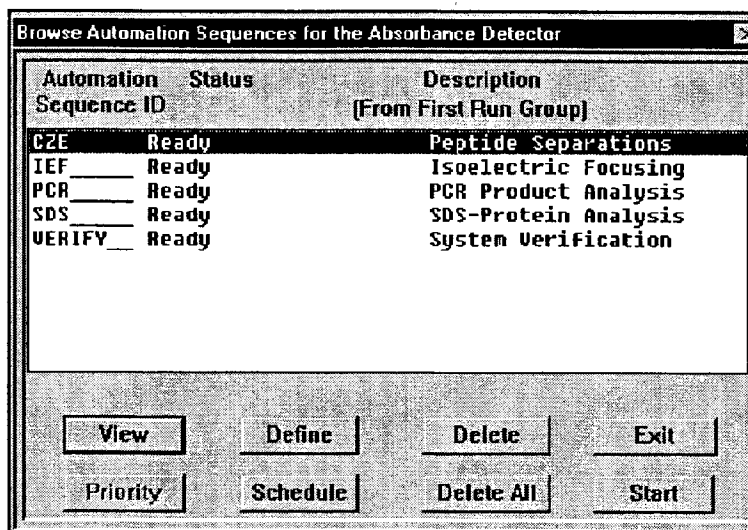


Figure 5-89 Browse Automation Sequences Screen

Viewing an Automation Sequence

If you wish to view your automation sequence, click the **View** button. Note that the View Automation Sequence screen shown in Figure 5-90 now lists all added run groups with their corresponding sample range, run sets, method ID's, run group status, waste positions, run group descriptions, and the configuration being used.

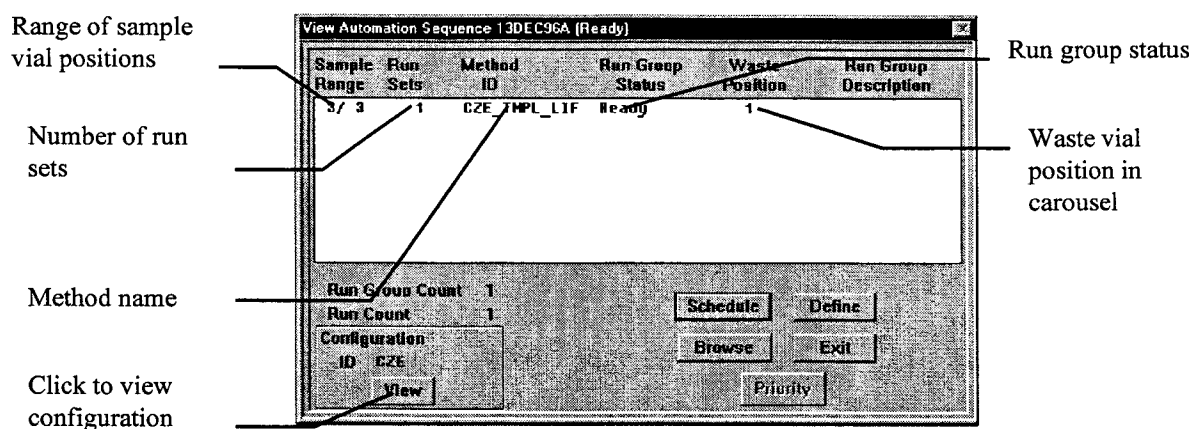


Figure 5-90 View Automation Sequence Screen

You may also view the configuration by clicking **View** at the bottom left corner of the View Automation Sequence screen.

Returning to the Main Screen

Clicking **Exit** in the View Automation screen brings you back to the main screen. Clicking **Browse** displays the Browse Automation Sequence screen. The **Priority** button allows you to schedule a priority automation sequence while another one is being executed. Priority automation sequences are initiated only after the current run is completed.

Reviewing/Editing Automation Sequence Details

Existing Automation Sequences cannot be edited in the literal sense. Instead, you may select an existing automation sequence as a template for defining a new automation sequence, and “edit” the new automation sequence before it is saved.

To edit a run group in an existing automation sequence, click **Define** on the View Automation Sequence screen. See Figure 5-91.

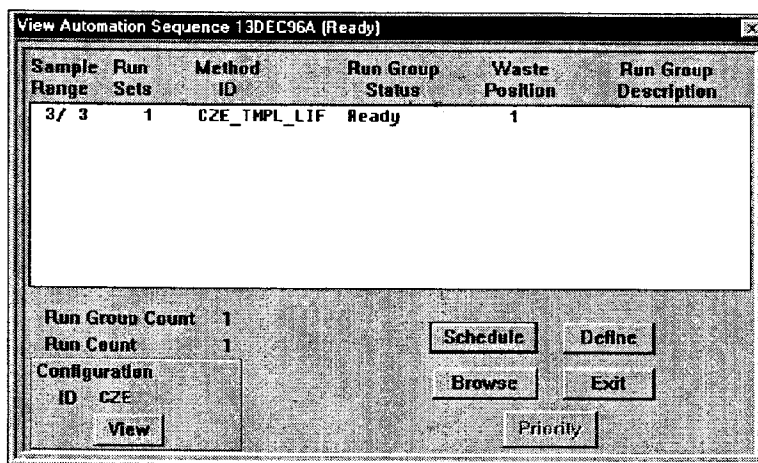


Figure 5-91 View Automation Sequence screen

This will take you back to Define New Automation Sequence ID and Select Configuration screen. Click **Continue** to bring up the Define Run Groups for Automation Sequence screen shown in Figure 5-92.

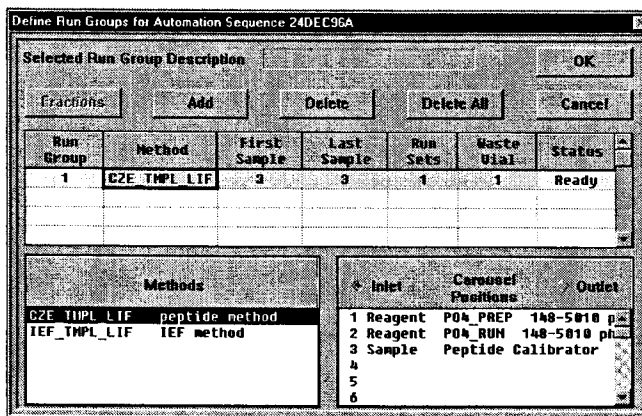


Figure 5-92 Define Run Groups for Automation Sequence screen

Scheduling an Automation Sequence

To schedule an automation sequence from the Browse Automation Sequence screen (see Figure 5-93), highlight the desired automation sequence, then click **Schedule**. Next click **Exit** to return to the main screen.

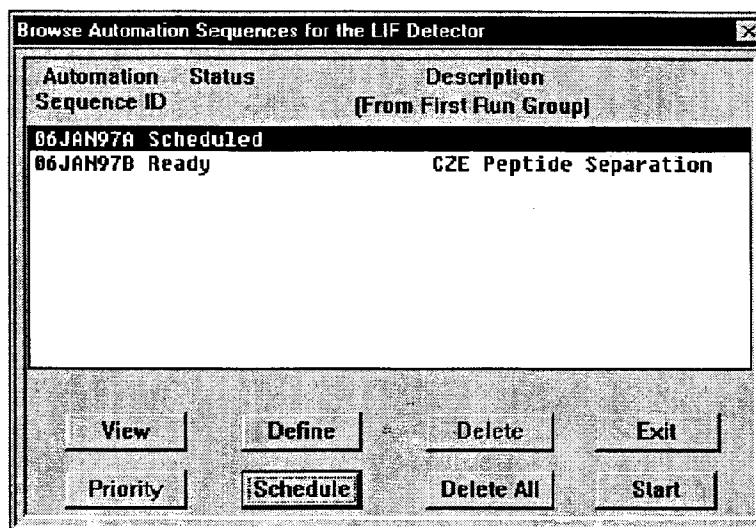


Figure 5-93 Browse Automation Sequence Screen—Scheduled

Alternatively, you can schedule your automation sequence from the View Automation Sequences screen. To access this screen, click the **View** button at the bottom of the Browse Automation Sequence screen, and then click **Schedule**.

You may program several automation sequences in advance, but only one can be Scheduled for execution. The rest remain in Ready status.

Scheduling Conflicts

If you attempt to start an automation sequence in Ready status while another automation sequence has been scheduled to start, you will get a warning message. Refer to Figure 5-94.

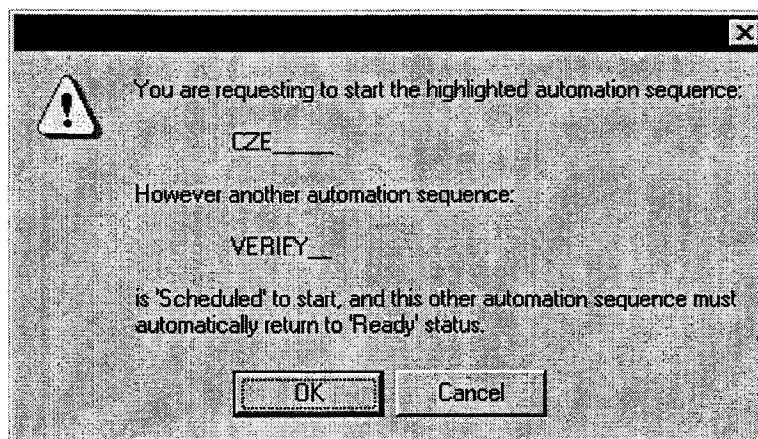


Figure 5-94 Scheduling Conflict Warning

Click **OK** or **Cancel** to continue.

Beginning an Automation Sequence

If there is already an automation sequence in Ready status, you may click **Schedule** and then click **Start** to begin the scheduled automation sequence from the Browse Automation Sequences screen (or from the Commands menu). Clicking the **Schedule** button changes the status of a “Scheduled” autosequence back to “Ready”. You can also start an automation sequence in Ready status without scheduling it by highlighting it and clicking the **Start** button. Clicking the **Delete All** button deletes **all** listed items except those that are scheduled, executing or interrupted. The Exit button lets you exit the Browse Automation Sequences screen.

Auto Integration

If you wish to integrate automatically after each run, prior to starting the next run in an automation sequence, you should create an Integration Method in the Integrator software. This software is used to define integration methods and process data obtained during a run. Refer to the BioFocus Integration manual for details. Use the appropriate integration parameters for each BioFocus Run Method of the automation sequence, or the Default1 Method Template will be applied automatically. Because the Default1 method template is supplied with integration turned off, you will get only a printout of the electropherogram at the end of each run.



Auto Integration Button

NOTE: Auto Integration is selected (on) if the toolbar button is “depressed” (also, under Commands, there will be a check mark next to this item).

Once you have created an integration method(s), scheduled an automation sequence, filled and loaded the vials in both carousels, installed a cartridge in the BioFocus instrument, and turned on the gas valve, you can go to the Commands menu and click **Start Scheduled Automation Sequence**. Or, you can start your automation sequence directly from the Browse Automation Sequences screen without scheduling it first, by simply clicking the **Start** button.

You may select integration on (or off) by clicking the Auto Integration button on the toolbar (or the Integration during Auto Seq item in the Commands menu). A report will be generated at the end of the next run and after all remaining runs in the sequence.

When running multiple wavelengths or high-speed scanning, only the shortest wavelength will be integrated between runs. Other wavelengths will require post-run analysis. Note that because autointegration is not performed when wavelength changes have been programmed, reports will not be generated at the end of such runs. These electropherograms can only be accessed through the Spectra (or View Electropherogram) post-run software.

Multitasking

You may review or define new configurations, methods, or automation sequences during a run without fraction collection simply by clicking the main menu bar. Periodically you may be preempted by the computer which is sending higher priority commands to advance the automation sequence which is under way. At these times neither the keyboard nor the mouse will be functional. Usually after a few seconds you will be able to resume programming. If you do multitasking, the Electropherogram will be hidden until you click Electropherogram on the top menu bar.

With the above exception, multitasking is not recommended during BioFocus operation. Although the Windows environment permits multiple applications to be opened simultaneously, only one application can be active at any one time, potentially compromising real-time data collection. This is due to the fact that Windows is not preemptive, that is, Windows cannot deactivate the currently active application—that application must voluntarily yield its active status before the computer can accept data from another application. This, combined with the responsiveness of current personal computers, limits data handling at two levels.

First, the computer may process data from the detector too slowly because of slow application switching when running Windows. The inadequate processing speed causes older data to remain in the receive register of the serial port. When a new data byte is received, the older data that was not forwarded to a higher level is overwritten and lost. This is an overflow error.

Secondly, at the higher level, the computer stores an amount of collected data in a circular software buffer of extremely limited size. This buffer must offload its contents to a permanent storage medium before it refills. A Windows software application is responsible for this offloading; however, it must be the active Windows application for it to perform this offloading service.

For example, consider a situation where a user is attempting to perform post-run integration while a BioFocus automation sequence is running. If integration of a file is not completed by the time the buffer needs to be vacated for the incoming automation sequence detector data, Windows will not be able to clear the buffer and valuable detector data points will be lost, effectively destroying the efficacy of the run.

Priority Runs

The **Priority** button allows you to schedule a priority automation sequence while another one is being executed. Priority automation sequences are initiated only after the current run is completed.

Duplicating/Rescheduling Completed, Ready or Aborted Automation Sequences

Completed, Ready, or Aborted automation sequences may be easily duplicated by highlighting the sequence in the Browse Automation Sequences screen and clicking **Define**. Click **Continue** in the next Define New Automation Sequence screen and **OK** in the next Define Run Groups for Automation Sequence screen. The automation sequence just defined will now appear Ready and can be Scheduled from the Browse Automation Sequence screen.

Indirect Detection

If you are performing indirect detection (negative peaks), before starting your automation sequence you should program an Integration Method from the Integration software with the Invert Negative Peaks option selected. If you use the Default1 method template, electropherograms will not be inverted.

Communication Errors

All communication errors during an automation sequence will have an automatic recovery attempt. An Event Log File will be created so that the user can link any errors with BFF files post-run. At the end of an Automation sequence the title of the closing message will be “completed” if any errors occurred.

Disk Space

Before starting an automation sequence the computer automatically calculates whether there is enough disk space to store all the data for the runs you have programmed. This calculation also includes approximately 3000 bytes/minute for integration of each run minute, whether you have or have not selected Integration Reports. Disk space during an automation sequence is filled up at the approximate rate of 0.35 megabytes/hour in single or multiple wavelength and 1.33 megabytes/hour in high-speed scanning mode. If you are monitoring many wavelengths, have long run times with many runs programmed, and have not recently downloaded older files from the hard disk, there may be insufficient disk space (a message will then appear).

If there is sufficient disk space and the cartridge has reached its proper temperature, you will observe the outlet carousel turn first to position the waste vial, followed by a rotation of the inlet carousel to the first purge reagent. A detector signal is displayed at the top of the screen. The trace shows an offset every time there is a purge or sample injection. Figure 5-95 shows a typical electropherogram display.

Automation Sequence Status

There is also a Current Operating Parameters box in the lower left corner. Information in this box tells you what part of the automation sequence is underway (Beginning, Retrying, Advancing, Prepping Cycle, Setting Cartridge Temperature, Injecting, Running, Mobilizing, Focusing, Problem, or Ending), as well as displaying current, voltage, sample position, cartridge and carousel temperature values. The last four digits in the number indicate which Run Group and run is underway.

Baseline Shifts

To eliminate baseline shifts at the beginning of a run, the baseline is automatically zeroed just prior to data collection. This results in an intentional, approximately 5-second delay in the start of data collection after the high voltage is turned on.

Electropherogram Display

When the actual run begins, the electropherogram will be displayed and collection of raw data is initiated. The run will not begin until the car-

tridge is within 1° C of the programmed temperature. When performing IEF, both the focusing and mobilization steps (raw data collection) will appear on the computer screen and be saved in a BFF file; however, the remote start signal to an external integrator will begin only at the time of mobilization.

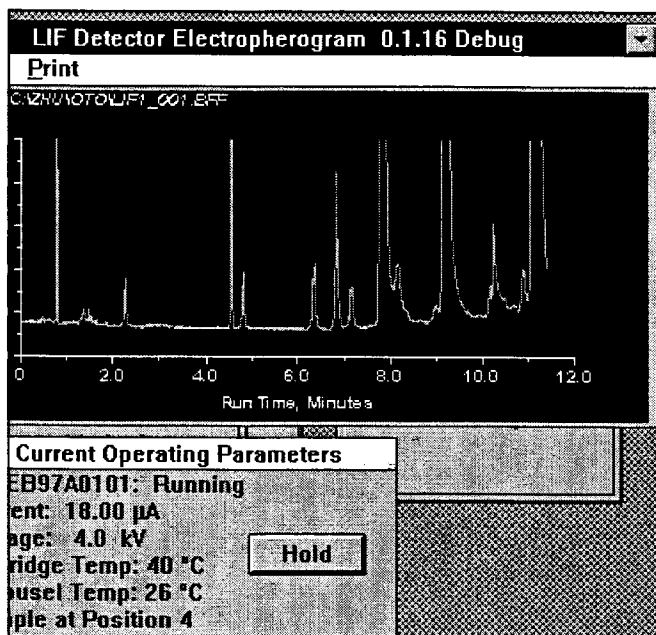


Figure 5-95 Live Electropherogram Display

To extend the life of your computer monitor's screen, you may dim it during long, unattended automation sequences. Turning off the monitor is *not* recommended, because it may cause power transients and communication problems with the computer.



CAUTION! Opening either the carousel or cartridge compartment door during an automation sequence interrupts the sequence and produces error messages.

Changing the Electropherogram Display

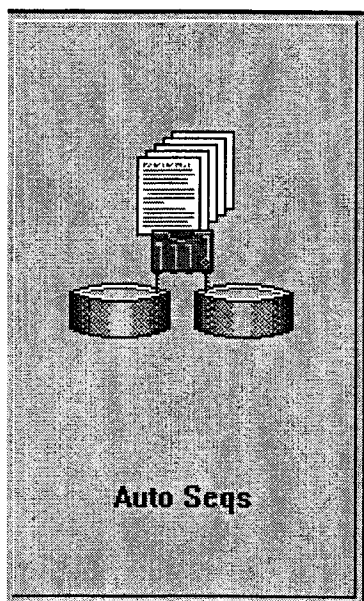
There is a menu under **Collect** in the upper left corner of the Electropherogram. By clicking **Display**, you can see the same screen which appears in the Display Parameters of the Detector section of Methods.

NOTE: To remove a pen assignment, you must click the pen and **none** in the wavelength selection box.

Changing the Display

You may to change the sensitivity scale, the baseline offset, pen color, and wavelength (provided that you have already programmed more than one wavelength in your Method) viewed during a run. Under the **Collect** menu, there are three other selections; **Event**, **Zero** and **Absorbances**. Clicking **Event** places an event mark on the electropherogram trace. Clicking **Zero** allows you to re-zero the baseline. Clicking **Absorbances** produces a box with numerical absorbance values for the wavelength(s) being monitored. To hide this box, click **Absorbances** again and the check mark disappears.

Priority Runs



Auto Sequences Button

NOTE: A Priority automation sequence can only be scheduled while another automation sequence is executing.

It is possible to interrupt an ongoing automation sequence to insert another run(s) by clicking **Auto Seq** on the menu bar (or the main screen Auto Seqs button) during a run. You can then define and schedule a new automation sequence for a Priority run or you can schedule an automation sequence that is already listed in Ready mode on this screen. Click the Ready automation sequence you wish to run, then click the Priority button. After you click the **Priority** button, the Priority Automation Sequence Options screen appears, showing a list of pause options. See Figure 5-96.

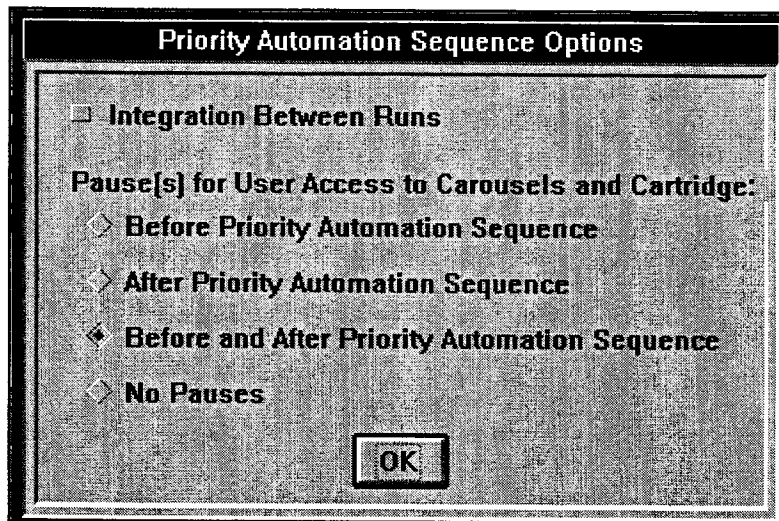


Figure 5-96 Priority Automation Sequence Options Screen

Before Priority... or After Priority... runs allow you to change the carousel temperature and cartridge (but the new cartridge use will not be recorded). You may also remove carousels, and add or remove sample, fraction, waste and reagent vials as necessary. Before and After Priority... allows you to make changes at both the beginning and end of the Priority sequence. If Priority samples and reagents are already in the carousels, No Pauses may be selected. If you selected After Priority . . . , after the priority automation sequence is completed, the Pause for Prior-

ity Automation Sequence screen shown in Figure 5-97 appears. If you wish, you can then reset carousel temperature before resuming suspended automation sequence by clicking Yes at the bottom of the message screen.

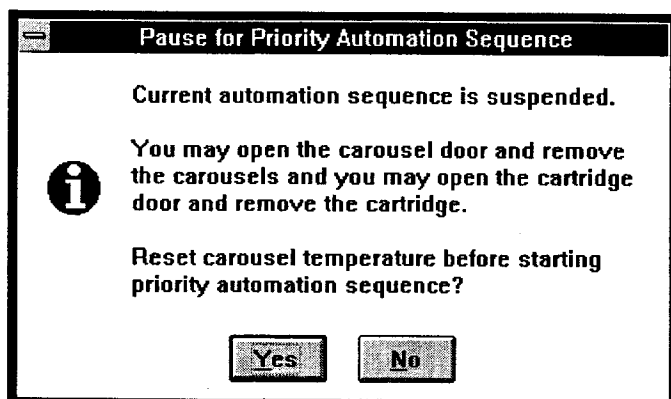


Figure 5-97 Pause for Priority Automation Sequence Screen

NOTE: Your selection for Priority Runs is independent of the selection you currently have set for the scheduled automation sequence.

Priority Options

The Priority Options screen shown in Figure 5-98 also provides an option for Integrator Reports to be printed between runs. When the original automation sequence resumes, reports will be printed out if you chose this for the original automation sequence. After you select the desired priority automation sequence options, click OK to schedule a priority run.

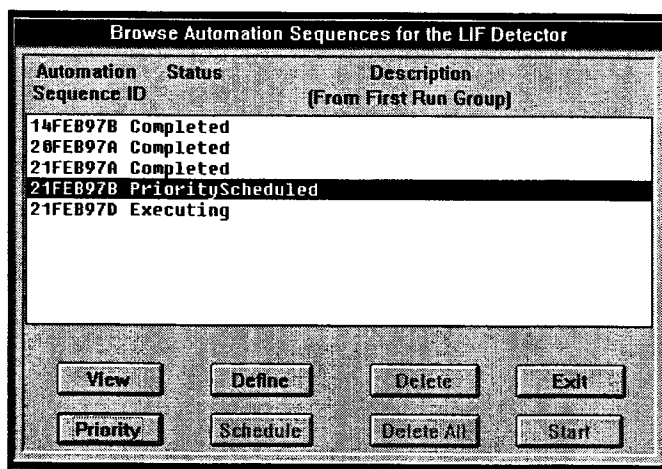


Figure 5-98 Browse Automation Sequences Screen-Priority Automation Sequence Scheduled

When an automation sequence is to be interrupted for a priority automation sequence, the current run is completed before the system pauses or initiates the priority automation sequence. A priority automation sequence can be a single sample with a single method, or a sequence of up to 99 runs with multiple methods and samples. Therefore, the Priority

Run function makes it possible to program another 99 runs for total of up to 198 runs, provided that the computer's hard disk has adequate storage space for all the new data. When you schedule a Priority run, the computer does not calculate the remaining disk space, so it is possible to run out of disk space before the Priority and original automation sequences are completed. This depends on the lengths of runs and number of wavelengths scanned. You should copy files to floppy disks or delete any nonessential data files prior to programming 198 runs.

Termination Options



STOP Button

You can terminate a single run, a run group, or an entire automation sequence by clicking the **STOP** button on the toolbar (or Terminate Automation Sequence Options under the Commands menu) and selecting the appropriate termination option on the dialog box shown in Figure 5-99. This command is active only when a run or an automation sequence is under way.

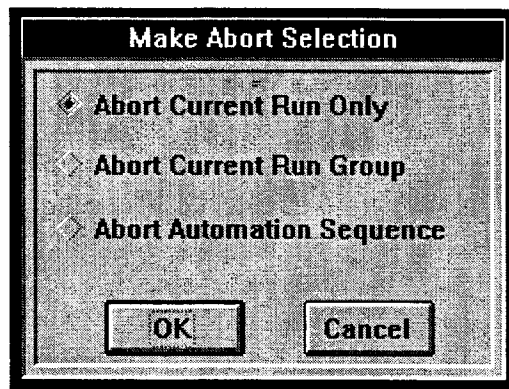


Figure 5-99 Abort Selection Box

If you abort an entire automation sequence, an Aborted message appears in the Browse Automation screen. If you abort only a Single Run or Run Group and there are no additional runs in the sequence, a Completed message will appear. All BFF (detection) files are saved in case you wish to analyze aborted runs later.

A user can program new setups, methods, and automation sequences while the detector is collecting data without the risk of losing data points. This is because the time intervals for these entries are so small. However, if a user chooses to delete large files such as an automation sequence containing 99 runs, this could take many seconds and cause the loss of detector data points.

End of an Automation Sequence

At the end of an automation sequence without anomalous events, a “Successful” message appears as shown in Figure 5-100. Any run-related information will be indicated by a “log” message at the end of the automation sequence name. If you like, you can save this automation sequence history for later review by clicking Yes.

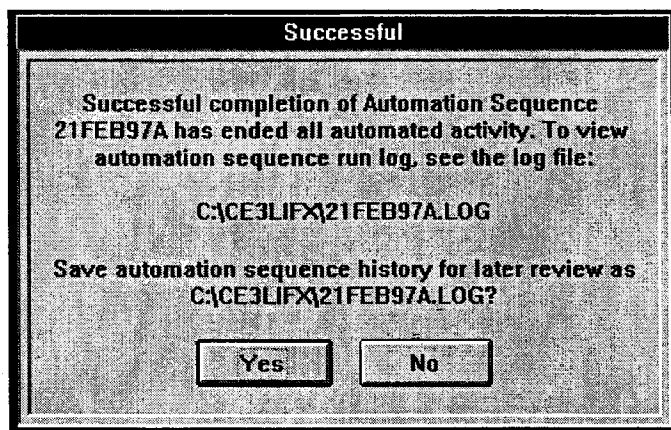


Figure 5-100 Successful End Of Automation Sequence Screen

If you selected inlet and outlet vials to remain in their upper positions with capillary and electrode tips immersed in the last run reagent (See “Vials Up at End of Run” on page 105), you must click **Yes** to lower the vials before attempting to remove either carousel. See Figure 5-101. Do not open either door before the hourglass symbol disappears and both carousels have rotated back to “home” position (otherwise, you will need to close the doors and push Reset on the rear panel to continue).

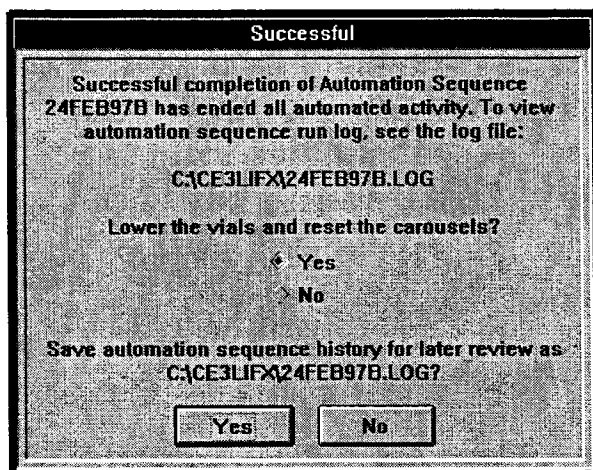


Figure 5-101 End Of Automation Sequence Screen When Vials Are Set To Remain In Upper Position

Log Message

A log message also appears whenever the user has chosen to terminate any portion of the automation sequence. In this case, a “Complete” message appears. Any errors and/or abort information will be indicated by a log message at the end of the automation sequence name.

Carousel Temperature

At the end of an automation sequence, the cartridge temperature will also return to ambient temperature. The carousel compartment remains at whatever temperature you have programmed until you turn the power off with the switch on the rear panel of the BioFocus instrument or until you reset it from the Carousel Temperature Control using the toolbar icon or the Commands menu. Do not leave the door open for long periods when the temperature is set lower than room temperature, because condensation will occur.

System Shutdown

To turn off the system and avoid corrupting the database, take the following steps:

For Windows 3.1 or 3.11:

Click **Exit** under the Commands menu. This closes the BioFocus software program. Next, Close the Windows Program Manager. The C> prompt appears.

For Windows 95:

Click the close button in the upper right corner of the screen to exit BioFocus software. Click the **Start** button and then the **Shut Down** button. If the **Shut down the computer** button is not selected, click it. Click **Yes** and wait for the “It’s now safe to turn off your computer” message to appear.

Turn off the power switch on the computer, then turn off the power switch on the BioFocus instrument. Finally, turn off the LIF Module if installed.

Other Commands—Menu Bar and Toolbar Options

The screen shown in Figure 5-102 illustrates the Main Screen Toolbar/Menu Bar command options available on a BioFocus 3000 system with carousel temperature control.

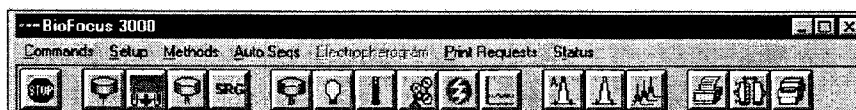


Figure 5-102 Main Screen Toolbar/Menu Bar

Vials Up at End of Run

This is an option that allows the vials to remain elevated at the end of an automation sequence or during Auto Integration to protect the capillary inlet and outlet from desiccation.



Vials Down Button



Vials Up Button

This command can be accessed either from the toolbar or the menu bar (toggle switch under Commands). The Vials Up (or Down) button on the toolbar has two positions: in the depressed position, vials will be up, and in the raised position, vials will be down at end of run (this change is also indicated by the arrow shown on the button). If you choose to leave the vials up, you will see the screen shown in Figure 5-103 at the completion of your automation sequence.

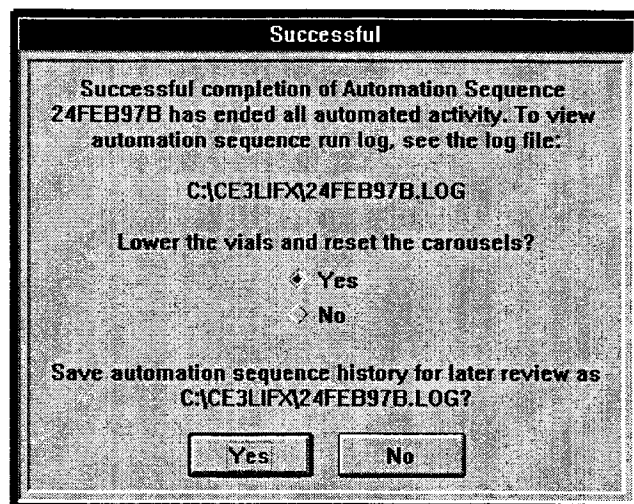


Figure 5-103 End Of Automation Sequence Screen When Vials Are Set To Remain In Upper Position

You have the option to lower the vials and reset the carousels by selecting the **Yes** button. If you select **Yes**, then vials will be automatically lowered and carousels will be brought to home position as soon as you **Exit** this screen. If you choose **No**, the vials will remain up and you will be able to program such functions as methods or automation sequences, but you will not have access to any of the items normally listed under **Commands**. Also, when you click **No**, the main screen toolbar disappears and is replaced with a text panel. Click the text panel to lower the vials and reset carousels (or click **Commands**, then on **Lower Vials and Reset Carousels**). Note that the toolbar now reappears.

This screen also allows you to save the automation sequence history for later review by clicking the **Yes** box at the bottom of the screen. To **Exit** this screen, you must click either the **Yes** or the **No** box at the bottom of the screen.

Reset All Communications



Reset communications button

NOTE: If for any reason you turn the BioFocus power switch off, it is essential that you **Reset** all Communications before resuming any operations

This command can be accessed from the toolbar button (or under **Commands**) to reinitiate (reset) communications between the computer and the BioFocus instrument, and between the computer and the detector. This command is inactive (grayed out) while a run or automation sequence is under way.

Detector Baseline Monitor (Absorbance Only)

This allows you to check the BioFocus detector for proper operation before an automation sequence is scheduled. This command is inactive (or grayed out) during a run. The detector baseline monitor allows you to view single or multiple wavelengths because it operates the detector in scanning mode.



Baseline Monitor

NOTE: You cannot monitor wavelengths between 366-369 nm and between 796-800 nm.

If you have a BioFocus 3000, click the **Baseline Monitor** button on the toolbar (or Detector Baseline Monitor under Commands). Then, select the Scanning Range (UV or Vis) from the Detector Baseline Monitor screen shown in Figure 5-104.

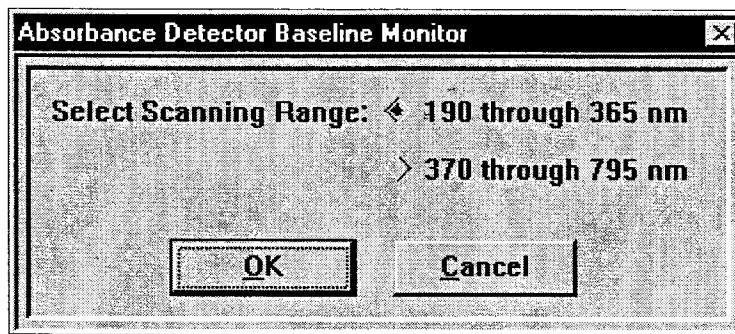


Figure 5-104 Baseline Monitor Screen BioFocus 3000

If you have a BioFocus 2000, the Detector Baseline Monitor screen shown in Figure 5-105 appears.

NOTE: Do not program runs in the visible range (wavelengths from 366 nm to 800 nm) without installing a tungsten lamp in your BioFocus 2000 instrument. No harm will occur, but the test will produce meaningless data.

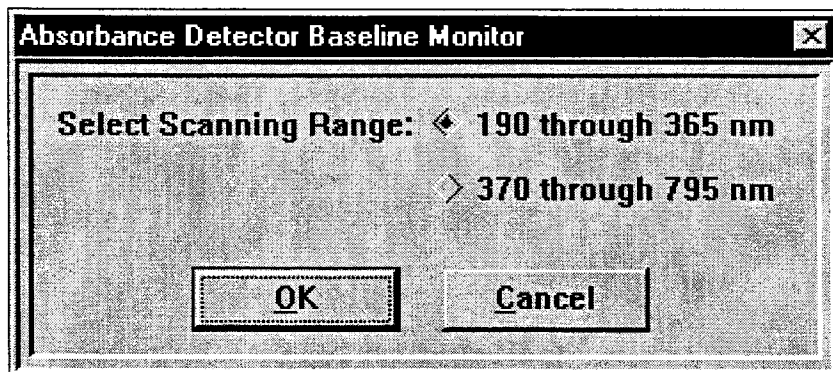


Figure 5-105 Baseline Monitor Screen BioFocus 2000

Enter the desired wavelength. With the BioFocus 2000, you can monitor wavelengths from 190 nm to 365 nm (UV). With an optional tungsten lamp, visible detection from 366 nm to 800 nm is also available.

When the detector trace is displayed on the screen, click **Display** under the **Collect** menu. See Figure 5-106. Select a pen color, a wavelength, and offset such as 0.02 AUFS and +10%. Click **OK**. Numeric absorbance values appear in a small window above the trace. When the Detector Baseline Monitor is on, the toolbar button will be in the depressed position (also, a toggle switch (check mark) appears in the Commands menu). Click the toolbar button to turn off the baseline monitor (check mark disappears). See “Changing the Electropherogram Display” on page 99 for more details.

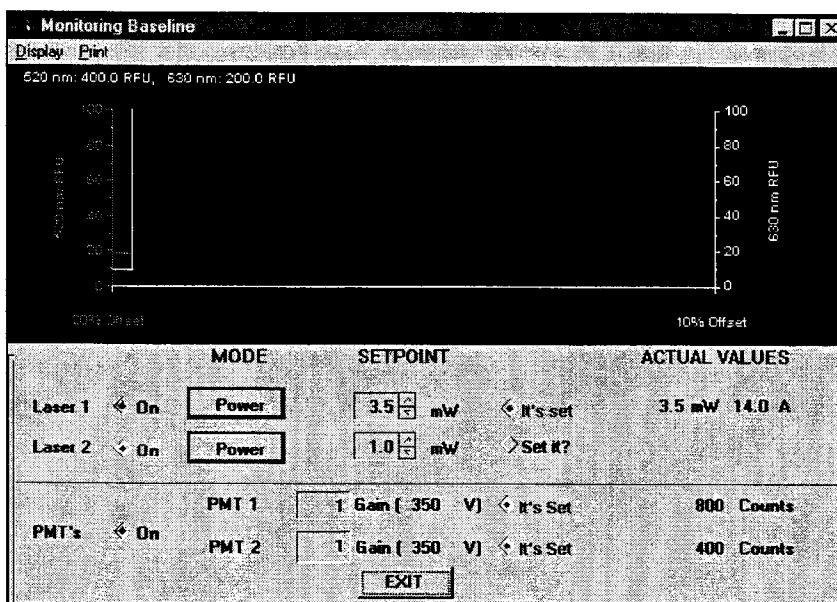


Figure 5-106 Monitoring Baseline

Diagnostics—Detector Autocalibrate (BioFocus 3000 Only)

The BioFocus self test may find an error in detector calibration. If it does, you will be instructed to run this autocalibrate command.

Reset Carousels



Reset Carousels Button

This command can be accessed from the toolbar button (or under Commands) to bring the carousels to their home position. This feature is inactive (grayed out) during a run or automation sequence.



Auto Integration Button

This button retains its setting even when the software is not running.

Integration During Auto Sequence

You can select this option by clicking the **Auto Integration** button on the toolbar (or **Integration During Auto seq** under the **Commands** menu). If you selected this command for an automation sequence, the toolbar button will be in the depressed position (also, under **Commands**, there will be a check mark next to this item). A report will be generated at the end of the next run and after all remaining runs in the sequence. You may also stop integration during autosequence by clicking the depressed toolbar button to bring it back to the raised position (also, under **Commands**, the check mark disappears).

Clean up the Database

This command initiates a database cleanup to check database integrity and to permanently remove all database files that were previously deleted. This command can be accessed only through the **Commands** menu.



Post-Run Integration Button

Post-run Integration

This command provides easy access to the Bio-Rad Integration software to create a new integration method, analyze data, or view an electropherogram. This command can be accessed both from the toolbar (using the **Post-Run Integration** button or the **Commands** menu).

About BioFocus 3000 (or BioFocus 2000)

This provides a simple description of the BioFocus software. It lists the ID (version) of the BioFocus program and the ID of the database currently residing in your computer memory. This feature is accessible only from the **Commands** menu.

Print Requests



Print Requests Button

If you wish to make hard copies of the Cartridge and Reagent Lists or any BioFocus Method and Configuration (including CZE and IEF Single Run Group Methods and Configurations), or Automation Sequence in the database library, you may do so through the **Print Requests** button on the toolbar (or the menu bar). This command is inactive during a run.

Click **Print Requests** and you get the screen shown in Figure 5-107.

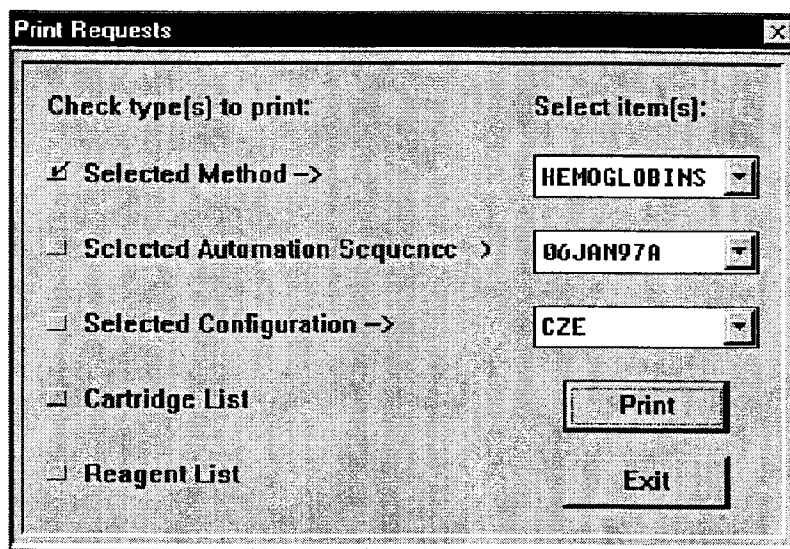


Figure 5-107 Print Requests Screen

If you wish to print a method, first check the appropriate box to the left of that item. See Figure 5-108.

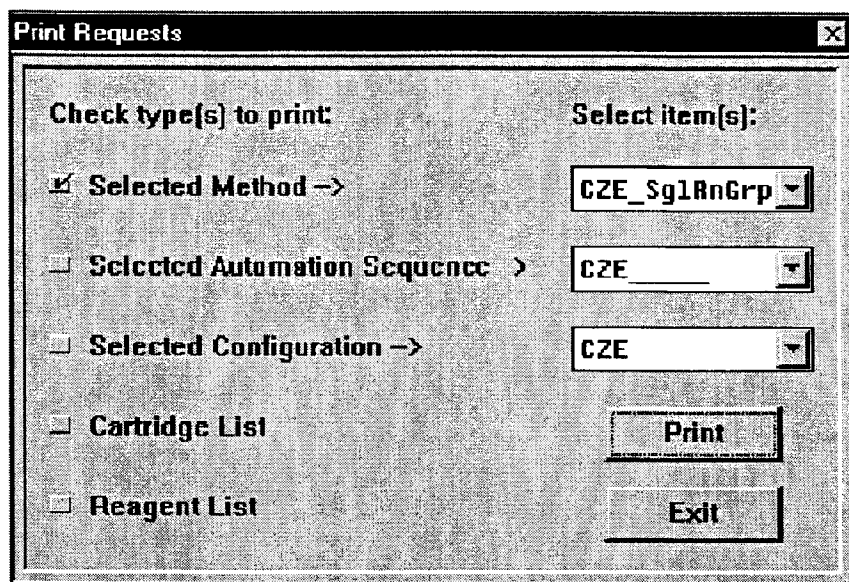


Figure 5-108 Print Requests Screen—Method Selected

Next, use the scroll bar to select the desired method from the list by highlighting it. The selected method now becomes the first entry under the Select item(s) box. Click **Print**.

You can check the five categories all at once. Make your selection from each category, and print.

Status



Status Button

There is a **Status** button on the main screen toolbar. You can use this to check on the status of BioFocus subsystems. You may also use status reporting to display revisions of the hardware embedded code in the BioFocus instrument. Refer to Figure 5-109. The status is only updated when you click the **Update** button.

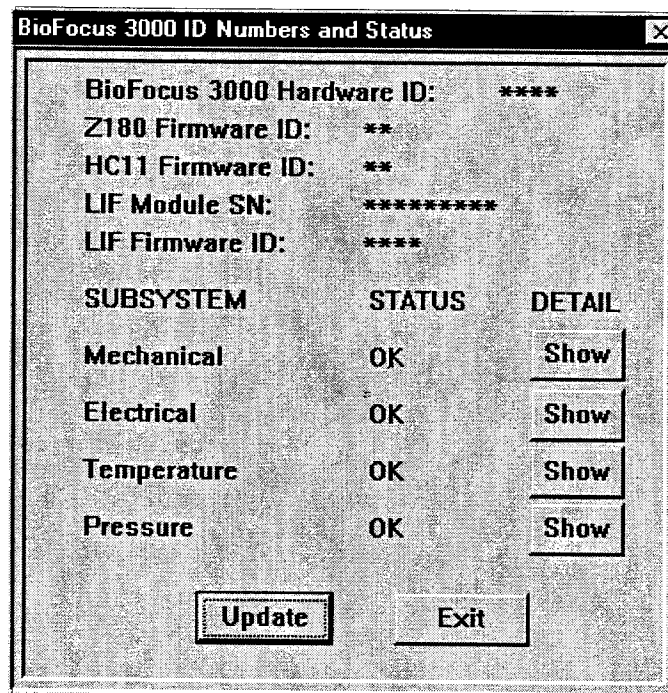


Figure 5-109 Status Screen

The first screen lists four subsystems you can check: Mechanical, Electrical, Temperature, and Pressure. Click **Show** for the function you wish to check.

The Mechanical screen (see Figure 5-110) shows which vial holders are raised (vertical) on both carousels as well as whether either compartment door is open.

MOTOR	POSITION	STATUS
Inlet Carousel	01	OK
Outlet Carousel	01	OK
Inlet Vertical	0.00 cm	OK
Outlet Vertical	0.00 cm	OK

DOOR	STATUS
Cartridge Compartment	Closed
Carousel Compartment	Closed

Figure 5-110 Mechanical Status Screen

The Electrical screen (see Figure 5-111) gives Mode, Range, Voltage, Current and Polarity information.

MODE	IN RANGE	
**	**	
CURRENT	POLARITY	VOLTAGE
0.00 μ A	**	0.0 kV

Figure 5-111 Electrical Status Screen

The Temperature screen (see Figure 5-112) shows current temperatures monitored at three different areas of the instrument.

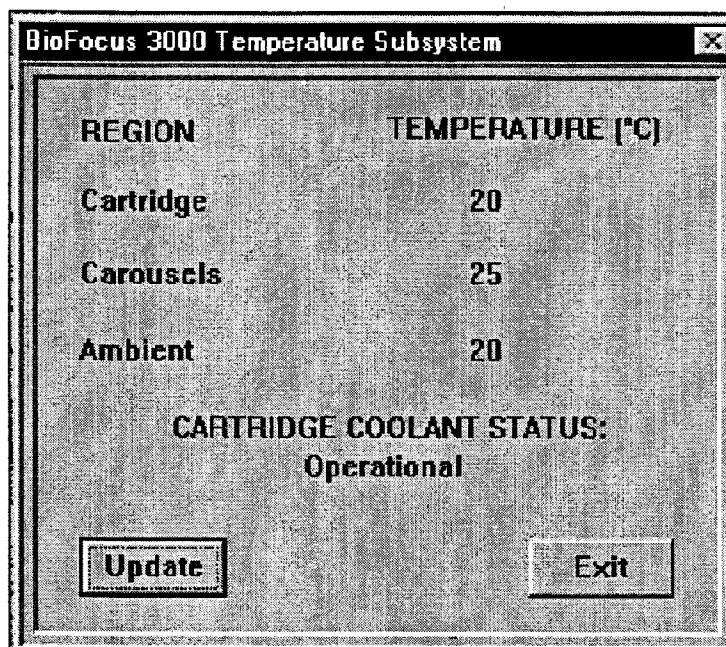


Figure 5-112 Temperature Status Screen

If you have a BioFocus system without carousel temperature control, then this Temperature screen (Figure 5-113) will be displayed.

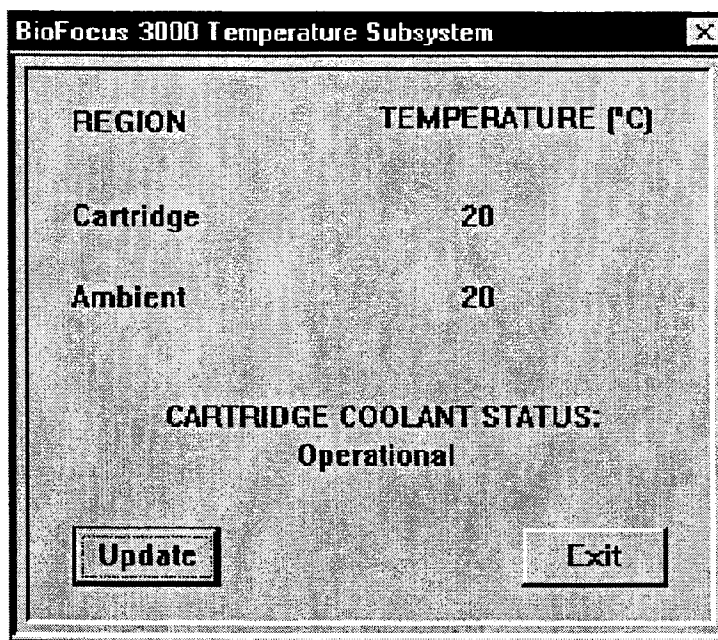


Figure 5-113 Temperature Screen Without Control

The Pressure screen (Figure 5-114) indicates whether high and low pressure purging are within range. If the high pressure is not on, the message indicates the status when the pressure was last turned on.

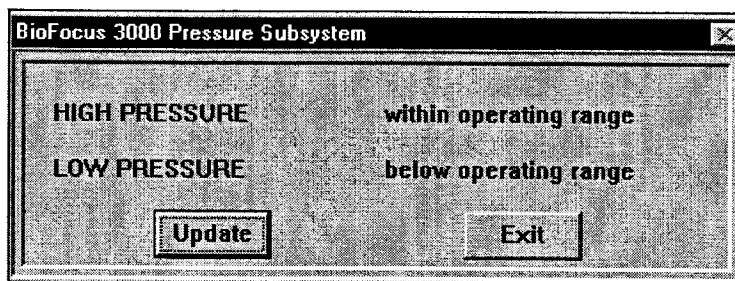


Figure 5-114 Pressure Status Screen

When you have finished checking any of the above screens, click **Exit** buttons until you return to the main screen, then click **Electropherogram** to bring the run back into view.



Electropherogram Button

View Electropherogram (BioFocus 2000 Only)

This command, accessible only from the toolbar, provides easy access to the View Electropherogram program. Use this command for displaying electropherograms and adding or subtracting one electropherogram from another.



Spectra Button

Spectra (BioFocus 3000)

This command, accessible only from the toolbar, provides easy access to the Spectra program for displaying electropherograms as single or multiple wavelength traces, for displaying spectra, and for analyzing spectra as first or second derivatives, additions, or differences.

Copying and Deleting Automation Files

Database records should be periodically archived to floppy disks or other media. Also, the database should be cleaned up. Once copied, the data files can be deleted from the hard disk to make space available for new Setup, Methods and Auto Seq files.

Individual records may be deleted within the BioFocus screens by clicking the file, then on the **Delete** button at the bottom of the screen. Obsolete records should be deleted in the following order: Auto Seq, Methods, Configuration, Cartridges, and Reagents. Template (STD) records may not be deleted. When all deletions are completed, click **Clean up the Database** under the Commands menu.

The following discussion assumes that the user accepted the default installation directory: CE3000 (or CE2000).

If you wish to save your databases for future reference or reloading, return to DOS (quit Windows) or go to the DOS shell in Windows 95, then insert a high density floppy disk in drive A (or B) and type:

```
Copy C:\CE3000\*.db a: (or b:)
```

then press the **Enter** key.

```
or Copy C:\CE2000\*.db a: (or b:)
```

then press the **Enter** key.

Then type:

```
Copy C:\CE3000\*.ndx a: (or b:)
```

then press the **Enter** key.

```
or Copy C:\CE2000\*.ndx a: (or b:)
```

then press the **Enter** key.

It is not possible to copy or delete individual *.db or *.ndx files without corrupting the database.



CAUTION! The database spans a number of disk files and individual files cannot be deleted outside the BioFocus screens. If you overwrite or delete specific database files, you will corrupt the database, effectively destroying it. Unfortunately, the symptoms of this corruption may not appear for some time.

It is possible to erase all databases at once by two methods. The easiest way to delete and replace all databases simultaneously is to put the BioFocus 3000 (or BioFocus 2000) installation disk into drive A, change to the CE3000 (or CE2000) subdirectory, and type the following at the C:\CE3000> (or C:\CE2000>) prompts:

```
a:newverdb
```

then press the **Enter** key.

Follow the instructions on the screen. This will conclude the procedure for an automatic replacement.

You may also delete all BioFocus databases by typing the following while in the CE3000 (or CE2000) subdirectory:

```
del *.db
```

then press the **Enter** key, and

```
del *.ndx
```

then press the **Enter** key.

Similarly, you can reload a database (saved on a floppy disk) from the CE3000 or CE2000 subdirectory by typing the following (assuming the floppy disk is put into the A drive):

```
copy a:*.db
```

then press the Enter key and

```
copy a:*.ndx
```

then press the **Enter** key.

You should also regularly copy and/or delete *.log and *.BFF files from the CE3000 (or CE2000) subdirectory. After 999 BFF files have been collected with the same method, the next BFF file with this method will overwrite BFF file 001 and the original data will be lost. Frequent removal of unnecessary files from the hard disk, particularly the larger high-speed scanning files, will make more disk space available for new data file storage.

Finally, you should defragment your hard disk periodically. Typical users should consider doing this about once a week.

Printing Computer Screens

If you wish to have hard copies of various BioFocus screens which you have programmed, you may do so through the Windows Write program. This is basically a simple process: first you must save the screen while in the BioFocus program, and then bring up the Windows program and click several items. Your computer printer should have at least 1.5 Mb of memory to print these screens. The following is an outline of the process:

Use your keyboard to save just the active window in the BioFocus program by pressing and holding down the **Alt** key and then pressing the **Print Screen** key, *or*, save all visible windows as they appear on the screen by pressing the **Print Screen** key alone.

Minimize the BioFocus program and maximize Windows Program Manager. Double-click the Write Program (it may be under Accessories) and maximize it.

Under the **Edit** menu, select **Paste**. Click **File**, click **Print**, then click **OK** to begin printer data transfer and printing.